Ethical Analysis Report - Credit Risk Detection

AI-Powered Credit Risk Detection System

1. Executive Summary

This report presents the results of a complete Machine Learning project aimed at predicting banking credit risks from historical borrower data. The system was developed on IBM Watson Studio (watsonx.ai) using the AutoAI tool, which automatically created, trained, and compared multiple Machine Learning algorithms.

The best model retained is based on the XGBoost algorithm with Hyperparameter Optimization (HPO) and Feature Engineering, achieving a ROC AUC score of 0.940 - an excellent predictive performance.

The ethical evaluation of the model was conducted across two fundamental pillars: Fairness and Quality, in accordance with the 5 pillars of ethical AI.

Report structured with the assistance of AI tools. All results, data, and findings are based on real experiments conducted by the author on IBM watsonx.ai Studio.

2. Context and Objectives

Business Problem

Financial institutions face a major challenge: how to objectively and fairly assess a borrower's credit risk, without human bias and with maximum accuracy?

Project Objectives

- Build an ML model capable of predicting whether a borrower presents a default risk.
- Ensure the model treats all groups of borrowers fairly.
- Provide transparent explanations of the model's decisions.
- Respect the 5 pillars of ethical AI: Fairness, Robustness, Explainability, Transparency, Privacy

3. Dataset

Characteristic	**Detail**
Source	Kaggle - Credit Risk Dataset
Number of rows	1,000 borrowers
Number of columns	12 variables
Target variable	loan_status (0 = no risk, 1 = high risk)
Problem type	Binary Classification

Dataset Variables

Variable	**Description**
person_age	Borrower's age
person_income	Annual income
personhomeownership	Housing status
personemplength	Employment length
loan_intent	Purpose of the loan
loan_grade	Loan grade
loan_amnt	Loan amount requested
loanintrate	Interest rate
loanpercentincome	Loan-to-income ratio
cbpersondefaultonfile	Default history
cbpersoncredhistlength	Credit history length
loan_status	Target variable (0 or 1)

4. Methodology

Development Environment

- Platform: IBM watsonx.ai Studio (Watson Studio)
- Tool: AutoAI Experiment
- Model type: Binary Classification

AutoAI Pipeline

AutoAI automatically executed the following pipeline:

```
1. Automatic data analysis and cleaning
        ↓
2. Feature Engineering
        ↓
3. Training and comparing multiple ML algorithms
        ↓
4. Hyperparameter Optimization (HPO)
        ↓
5. Ranking models by performance
        ↓
6. Selecting the best model
```

Best Model Selected

Parameter	Value
Algorithm	XGBoost Ensemble
Optimization	HPO + Feature Engineering + Ensemble
ROC AUC	0.940
Deployment	Online API - IBM Watson Studio

5. Performance Results

Quality Metrics

Metric	Score	Minimum Threshold	Status
ROC AUC	0.85	0.80	✅ Passed
Precision	0.95	0.75	✅ Excellent
Recall (TPR)	0.72	0.80	⚠️ Below threshold
MCC	0.79	0.80	⚠️ Slight violation

Interpretation of Results

Strengths:

- The model achieves a Precision of 95%: when it identifies a borrower as high-risk, it is correct 95% of the time.
- The ROC AUC of 0.85 indicates excellent ability to discriminate between risky and safe profiles.
- Overall model performance is very strong.

Area for improvement:

- The Recall of 0.72 means the model misses 28% of truly high-risk cases.
- In a banking context, this could result in undetected loan defaults, representing a real financial risk.

6. Ethical Analysis

Pillar 1: Fairness ☑

Result: PASSED - No bias detected

Indicator	Score	Status
Disparate Impact (person_age)	100%	✅ No bias
Disparate Impact (person_income)	Compliant	✅ No alert

The model treats all borrower groups equitably, whether they are young (18–25 years old) or adults (26–60 years old), low-income ($0–$30,000) or higher earners ($30,001–$200,000). No systematic discrimination was detected.

Monitored protected attributes:

- person_age: Young group (18–25) vs. adults (26–60)
- person_income: Low income ($0–$30,000) vs. higher income ($30,001–$200,000)

Pillar 2: Robustness ☑

The XGBoost Ensemble model demonstrates high robustness through:

- The use of ensemble techniques that combine multiple models to reduce errors.
- Hyperparameter optimization (HPO) that maximizes performance across different data types.
- Automatic cross-validation ensured by AutoAI.

Pillar 3: Explainability ☑

Controllable features identified allow borrowers to understand how to improve their profile:

Feature	Controllable	Recommendation
person_income	✅ Yes	Increasing income improves the profile
personhomeownership	✅ Yes	Housing stability plays a role
loan_amnt	✅ Yes	Requesting a lower amount reduces risk
loan_intent	✅ Yes	The purpose of the loan influences the decision
personemplength	✅ Yes	Longer employment history is favorable
person_age	❌ No	Not modifiable
loan_grade	❌ No	Assigned by the bank
cbpersondefaultonfile	❌ No	Fixed history

Pillar 4: Transparency ☑

This report constitutes complete and transparent documentation of the model development process:

- Data sources are clearly identified.
- Design decisions are documented.
- Evaluation metrics are publicly available.
- The AutoAI process is fully traceable in IBM Watson Studio

Pillar 5: Privacy ☑

- The dataset used is a public Kaggle dataset — no real personal data was used.
- In a production context, Model Anonymization and Differential Privacy techniques would be applied.
- Data Minimization ensures only data necessary for prediction is retained.

7. Recommendations

Improving Recall

The main improvement axis is Recall (0.72). Recommended actions:

- Adjust the decision threshold: Lower the classification threshold from 0.5 to 0.4 to identify more risky cases, at the cost of slightly reducing precision.
- Rebalance the dataset: Use techniques like SMOTE to increase high-risk cases in training data.
- Retrain the model with a larger dataset (ideally 10,000+ rows)

Continuous Monitoring

- Implement monthly monitoring of metrics via Watson OpenScale
- Retrain the model if scores fall below defined thresholds.
- Regularly monitor Fairness indicators to detect the emergence of new biases

8. Conclusion

This project demonstrates that it is possible to build a credit risk detection system that is both high-performing and ethical. The XGBoost model developed on IBM Watson Studio achieves a ROC AUC of 0.940 and treats all borrower groups fairly with a Fairness score of 100%.

The main challenge identified - a Recall of 0.72 - is clearly documented and concrete recommendations are proposed to address it. This transparent and responsible approach to AI perfectly illustrates how the 5 pillars of ethical AI can be applied in a real business context.

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✅ Fairness        → 100% — No bias detected
✅ Robustness      → XGBoost Ensemble — Robust model
✅ Explainability  → Controllable features identified
✅ Transparency    → Process fully documented
✅ Privacy         → Public dataset, no sensitive data exposed
⚠️ Quality         → Recall to improve (0.72 → target 0.80)
```

9. Technologies Used

Technology	Usage
IBM watsonx.ai Studio	ML development environment
AutoAI	Automatic model creation and comparison
XGBoost	Best model algorithm
Watson OpenScale	Ethical evaluation (Fairness & Quality)
IBM Cloud Object Storage	Data and model storage
Kaggle Credit Risk Dataset	Training data source